

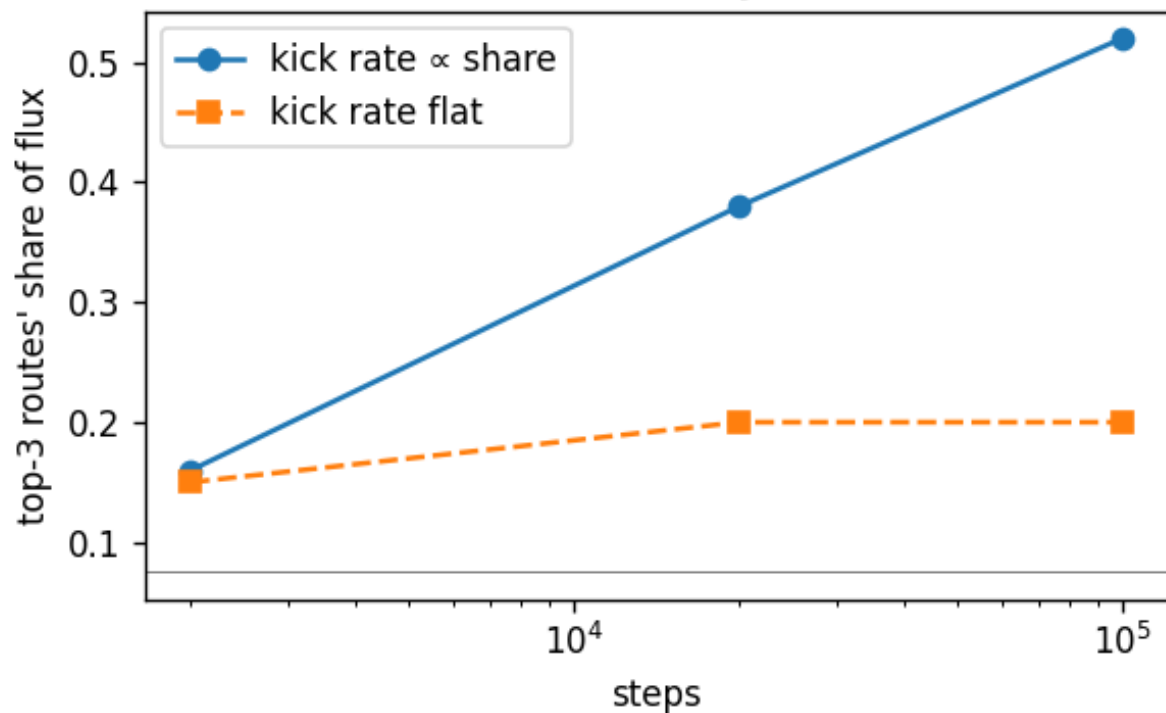
Natural selection as physics: each experiment in one sentence, with its figure

Core (the theorem and its checks)

1. Concentration, forty routes.

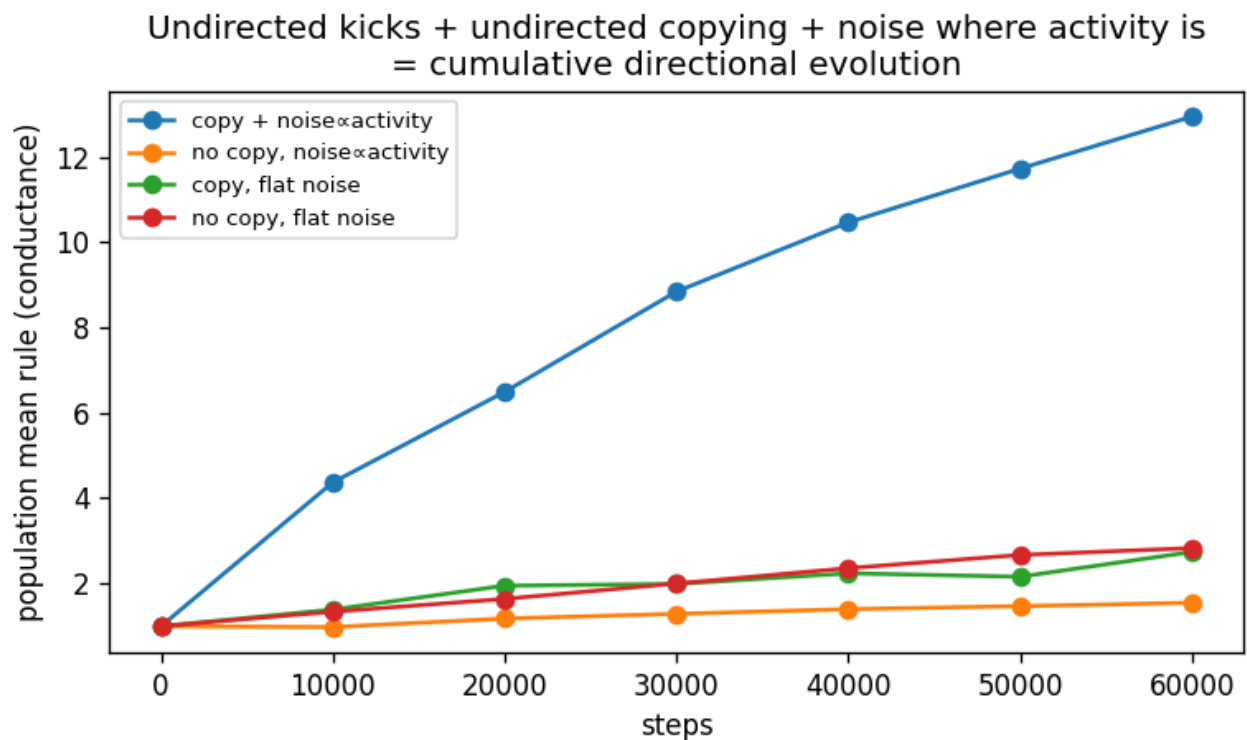
When channels sharing a fixed flow are disturbed in proportion to how much they carry, the flow ends up concentrated on a few of them, and when they are disturbed equally it doesn't.

undirected kicks: flux concentrates only when noise scales wi



2. Heredity, copying climb.

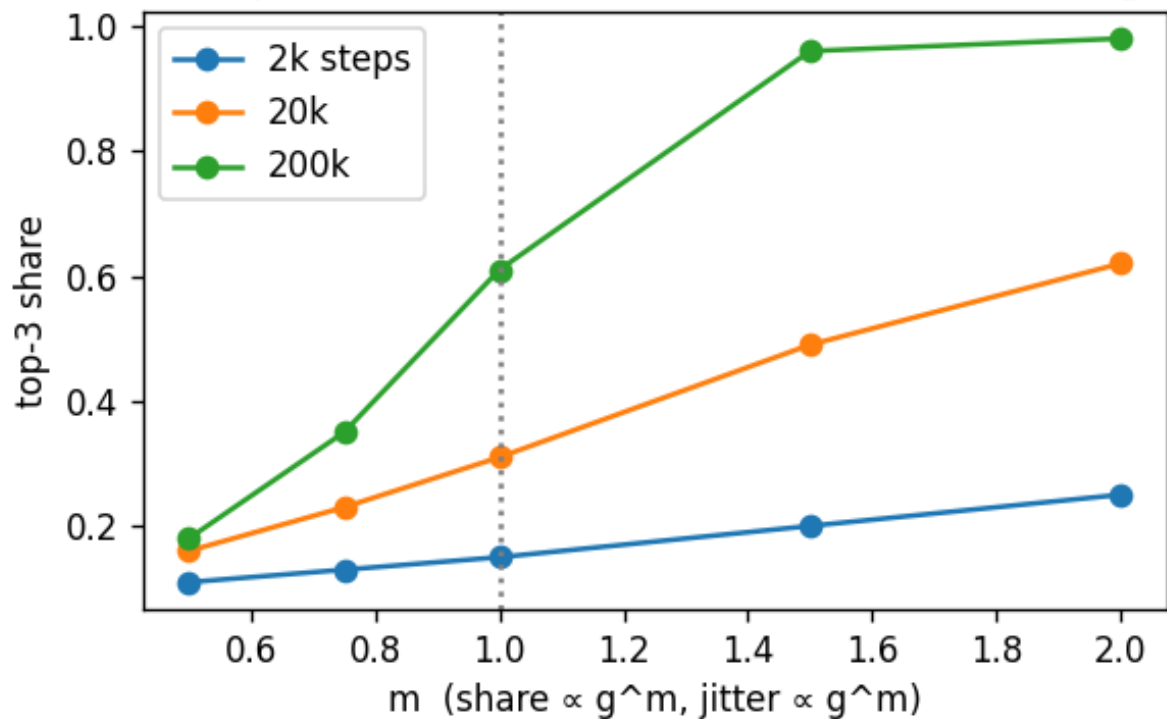
When busy channels also copy their state to neighbours more often, the whole population's average state rises steadily, and without either the copying or the activity-scaled disturbance it goes flat.



3. Exponent sweep on the toy line.

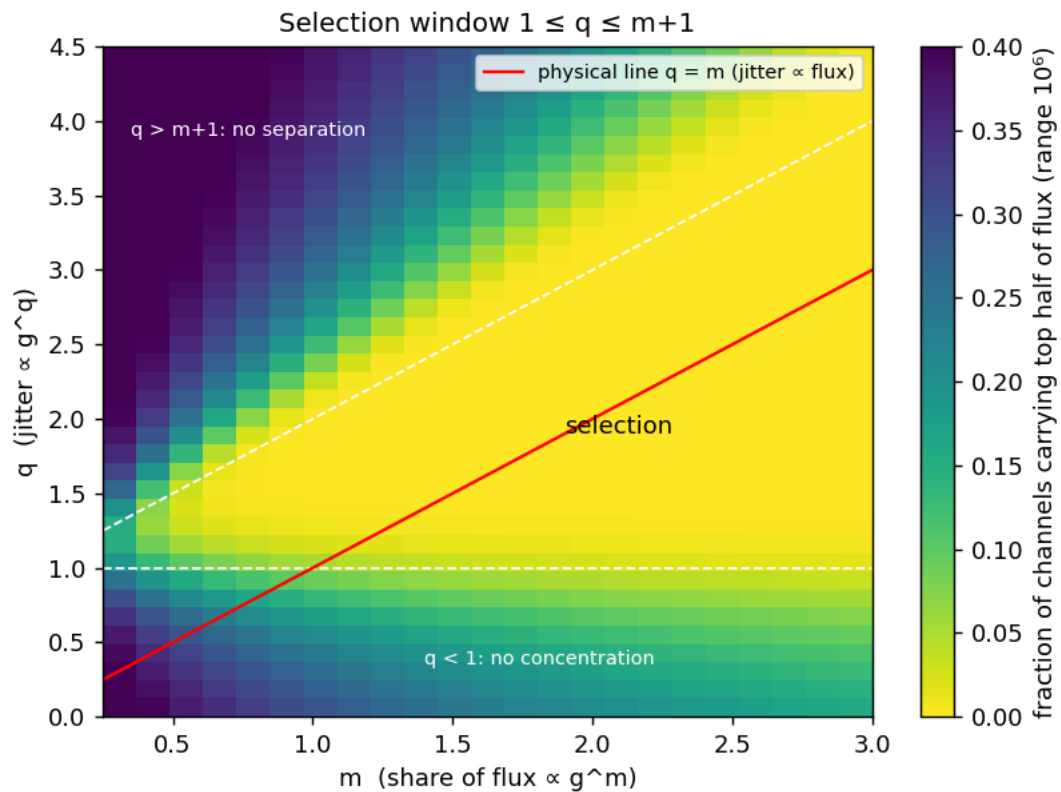
When share and disturbance both rise with state to the same power, concentration appears only once that power reaches one, slowly at exactly one and fast above it.

On the line $q = m$: no concentration below $m = 1$, marginal



4. Selection window, exact law.

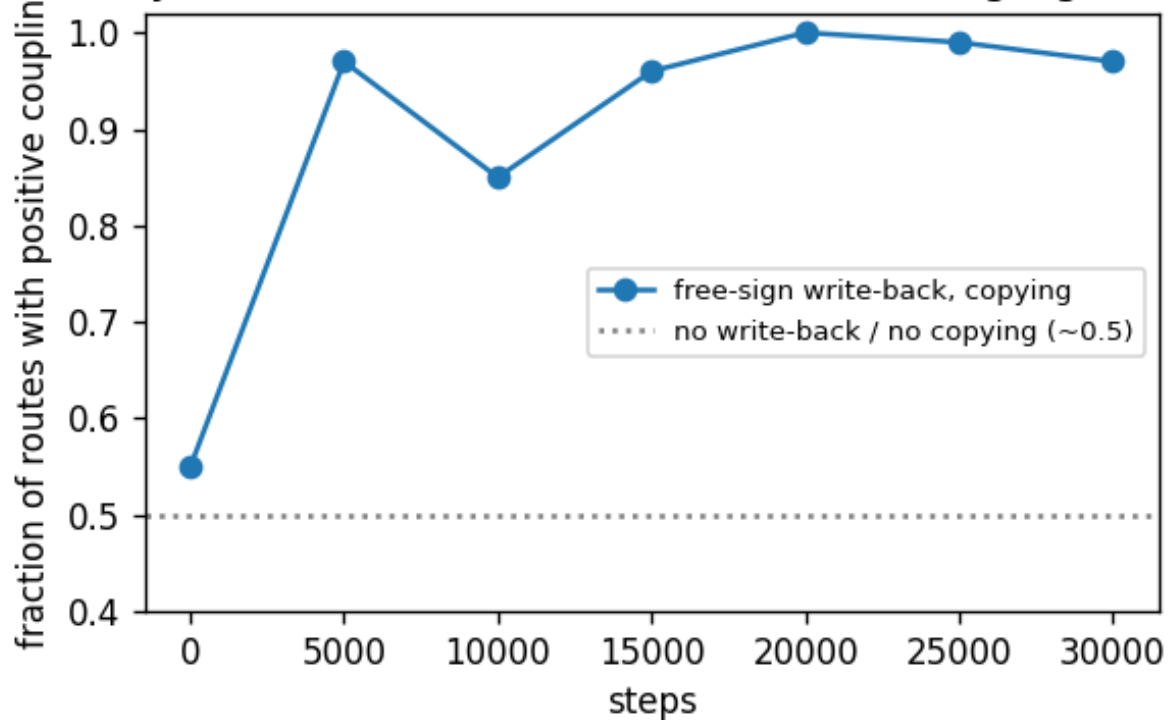
Concentration of flow onto a vanishing few happens only when disturbance grows with state at least linearly and no more than one power faster than flow does, and outside that window it never happens however long you wait.



5. Write-back, free sign.

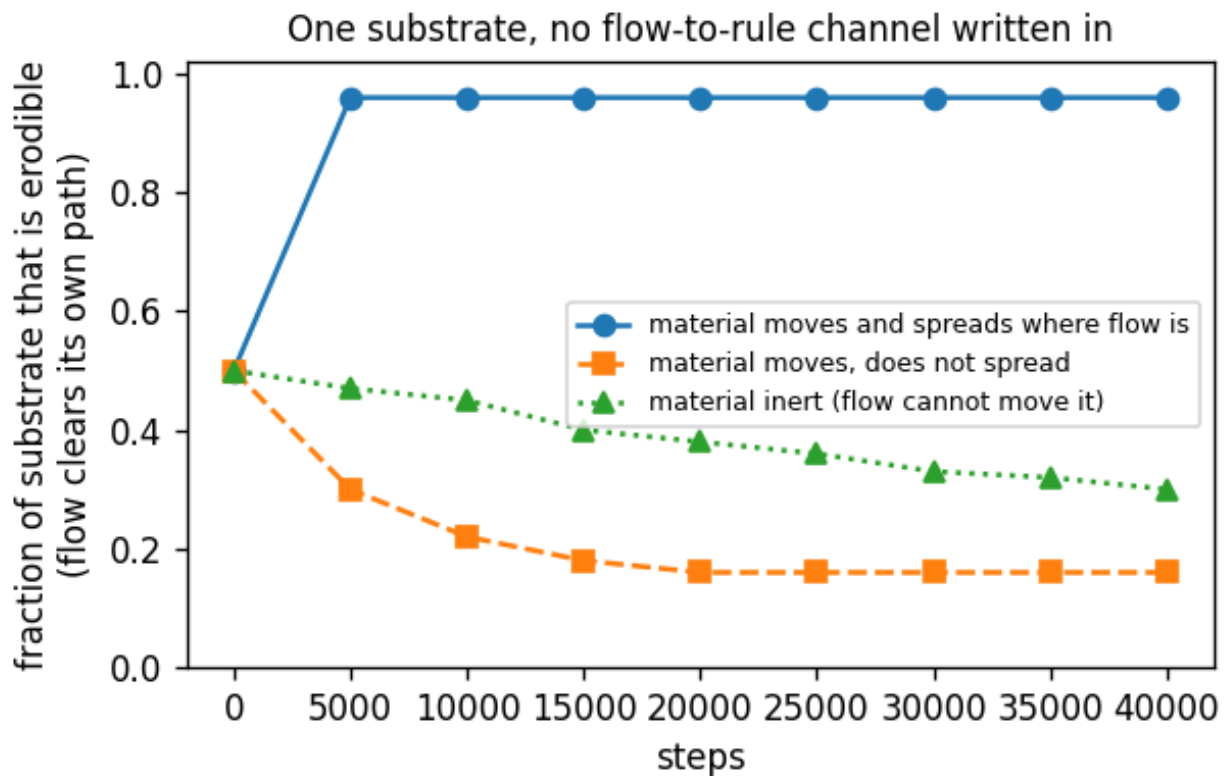
When each channel's own flow can rewrite its state with a sign that starts random, the channels whose flow reinforces them take over the population, and without copying they win the flow but don't spread.

Given any flow-to-rule channel, the self-reinforcing sign is se



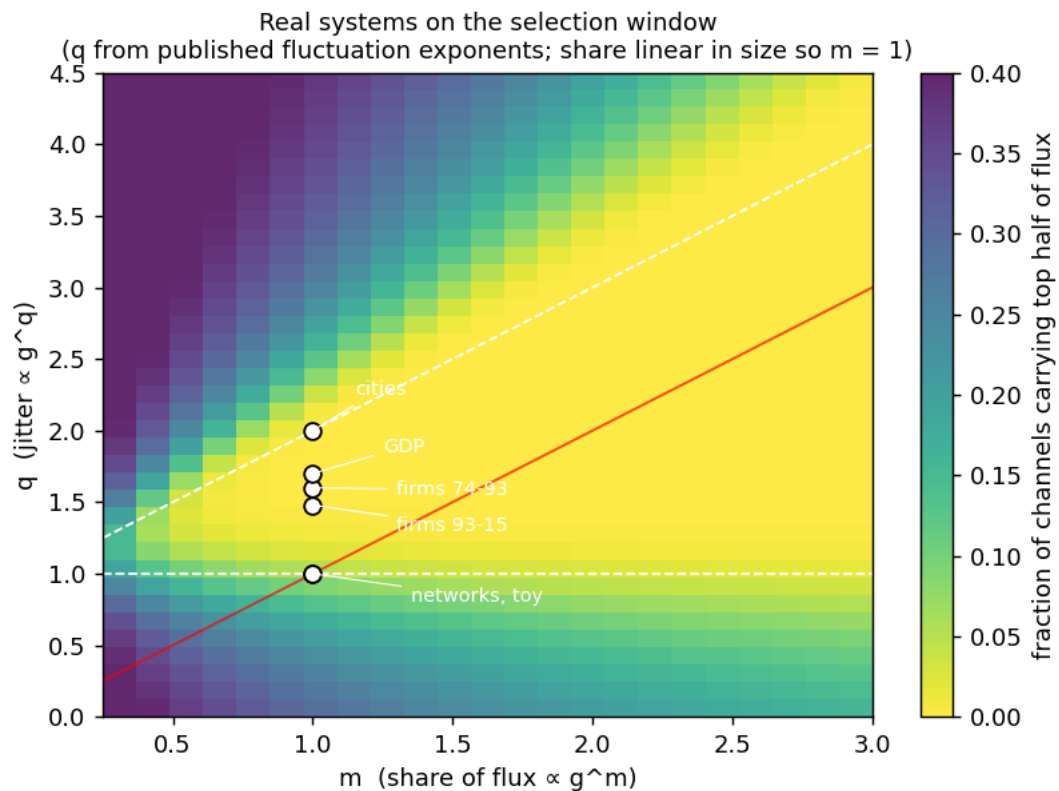
6. One substrate, no channel drawn.

When material obstructs flow, flow moves material according to the material's own nature, and material spreads where flow is, the kind that clears its own path takes over the substrate without any rule connecting flow to material being written in.



7. q from published exponents.

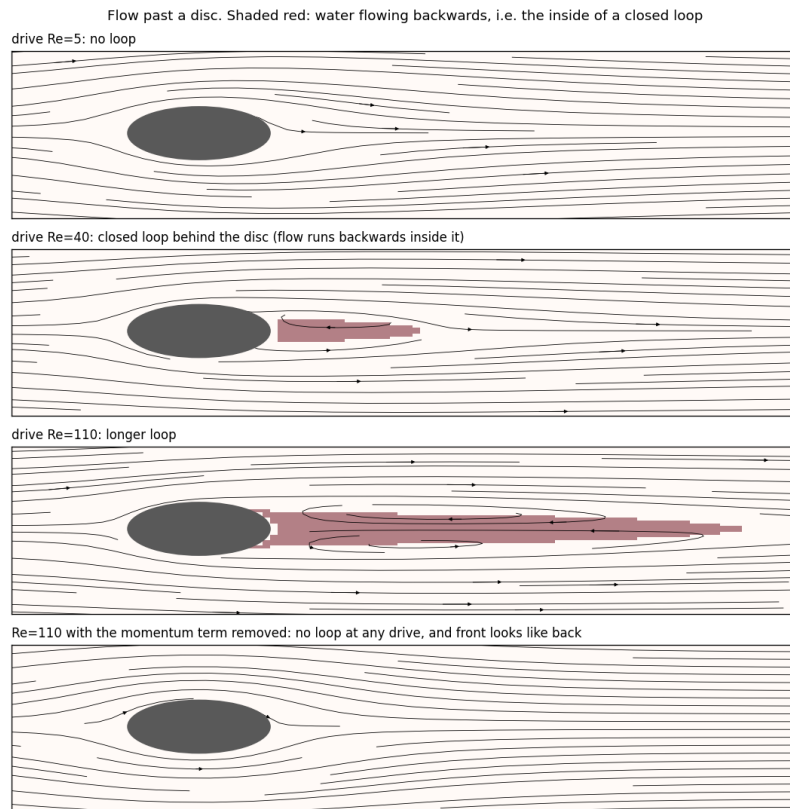
Where fluctuation size versus size has been measured, cities, firms, GDP and growing networks all land inside the predicted window, and the firm case is nearly but not exactly right, with the gap explainable by the upper tail not yet being stationary.



Boundaries (what linear systems cannot do)

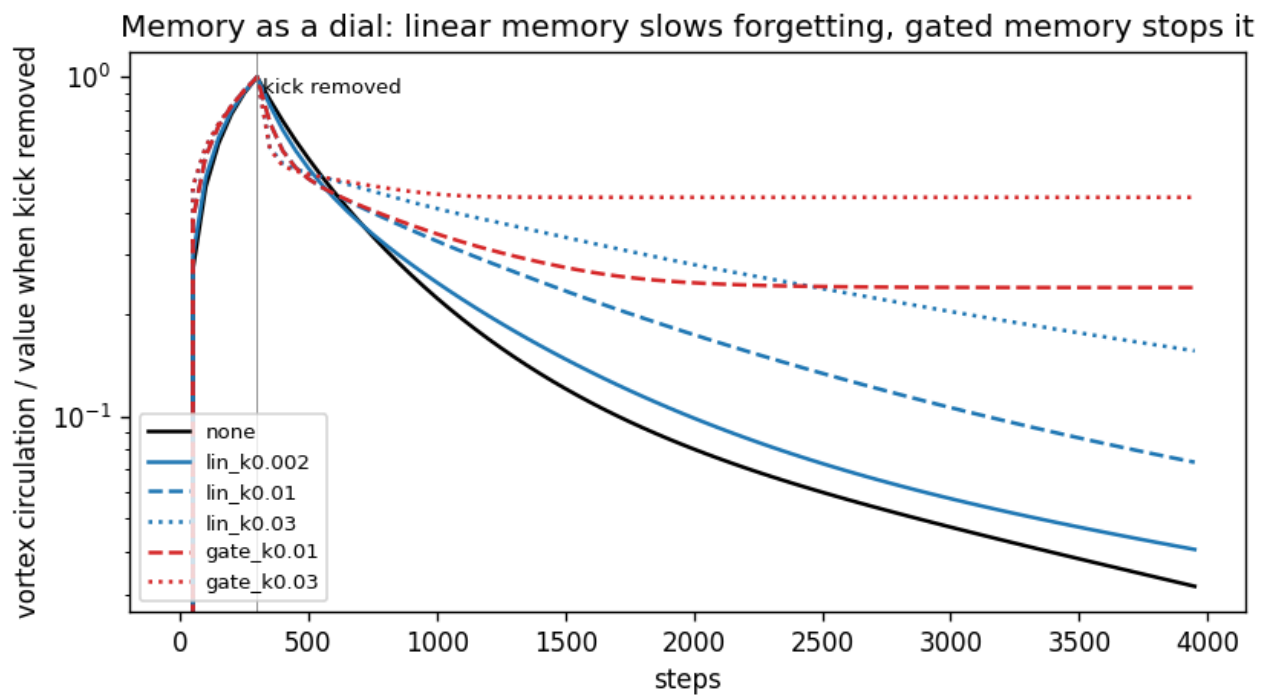
8. Lattice flow past a disc.

When a flow is driven past an obstacle, a closed loop appears above a threshold and grows with drive, and with the momentum term removed no loop forms at any drive.



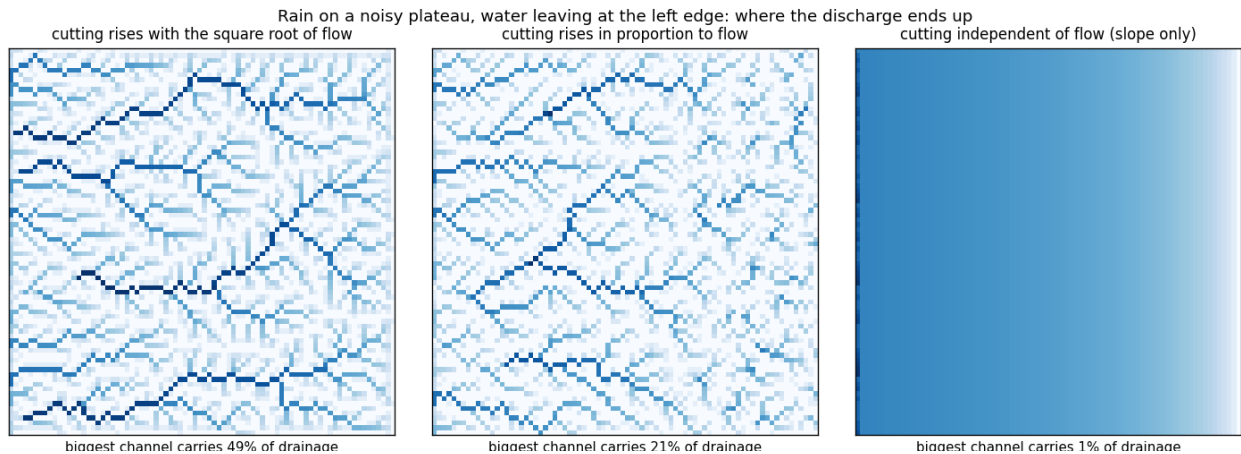
9. Memory gate versus dial.

A memory that is continuously rewritten only forgets more slowly however strong you make it, and a memory that writes only changes above a threshold holds a pattern indefinitely.



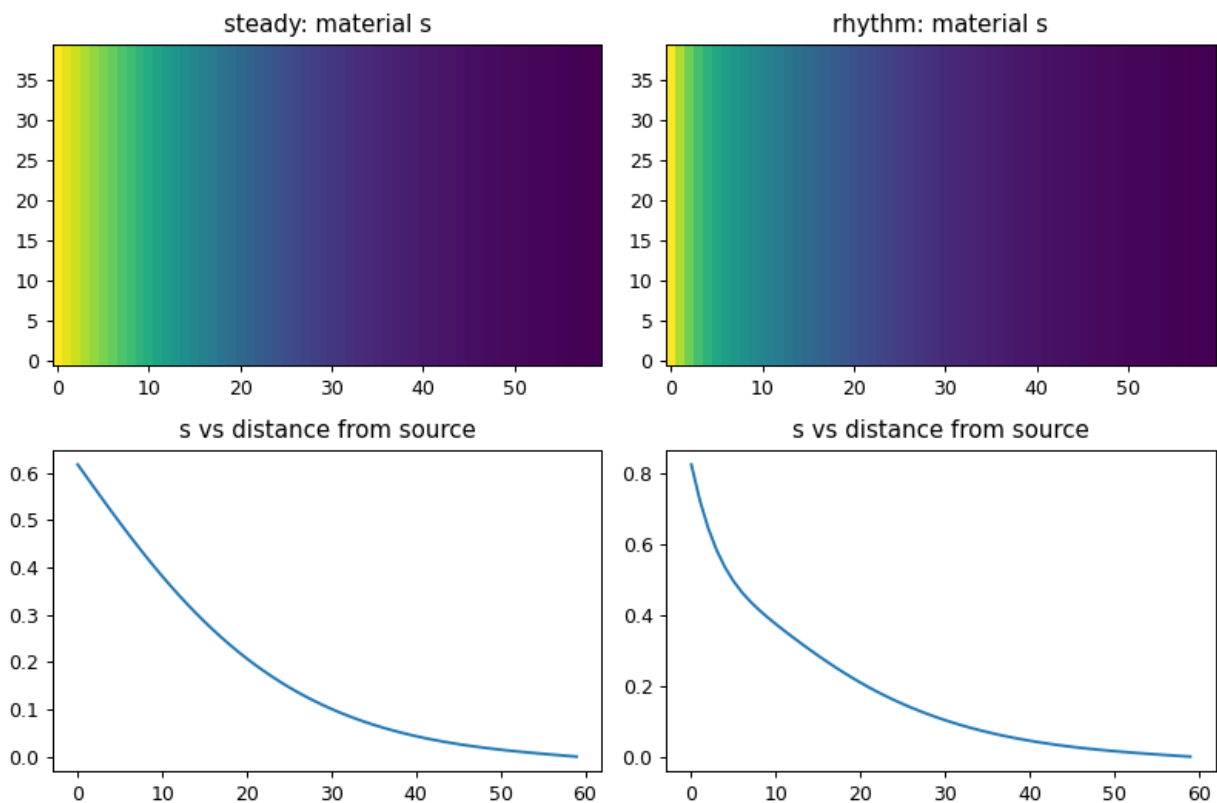
10. Plateau channel network.

When cutting rises with the water passing, rain on a noisy plateau carves itself a channel network in which a few channels carry most of the drainage, and when cutting depends on slope alone it erodes as a smooth sheet.



11. Blinking gradient.

When a gradient is switched on and off over a linear medium, storage earns a little extra throughput but no pattern ever forms.



12. River-bed hybrid.

When a loop forms behind a bump, removing the flow and keeping the bump lets the loop regrow, and keeping the flow and removing the bump kills it for good, so the memory lives in the slow layer and not in the flow.

Where does the pattern live?

